Exercise 4 Joins, UNION, Filtering, and Aggregates

Table 1: Employees

EmployeeID	FirstName	LastName	Department	Salary	City	Age
1	John	Doe	IT	70000	New York	35
2	Alice	Smith	HR	60000	Los Angeles	40
3	Bob	Johnson	Finance	75000	Toronto	45
4	David	Brown	IT	72000	London	30
5	Emma	Wilson	Sales	65000	Sydney	28
6	Michael	Clark	Finance	80000	New York	50

Table 2: Projects

ProjectID	EmployeeID	ProjectName	Budget	StartDate	Status
101	1	AI Development	100000	2024-01-10	Completed
102	2	Employee Training	50000	2024-02-15	Ongoing
103	1	Cybersecurity Audit	75000	2024-03-05	Pending
104	3	Financial Analysis	90000	2024-04-12	Ongoing
105	5	Market Expansion	65000	2024-05-20	Completed
106	6	Risk Management	80000	2024-06-10	Pending

1 Retrieve all employees and their assigned projects using an INNER JOIN. Return: EmployeeID, FirstName, LastName, Department, Salary, ProjectID, ProjectName, Budget, Status.

```
SELECT A.EmployeeID, FirstName, LastName, Department,
        Salary, `ProjectID`, `ProjectName`, Budget, Status
FROM `workspace`.`default`.`table-1-employees` AS A
INNER JOIN `workspace`.`default`.`table-2-project` AS B
ON `A`.`EmployeeID` = `B`.`EmployeeID`;
```

Employee ID	FirstName	LastName	Department	Salary	Project ID	ProjectName
1	John	Doe	IT	70000	101	AI Development
2	Alic	Smith	HR	60000	102	Employee Training
1	John	Doe	IT	70000	103	Cybersecurity Audit
3	Bob	Johnson	Finance	75000	104	Financial Analysis
5	Emma	Wilson	Sales	65000	105	Market Expansion
6	Michael	Clark	Finance	80000	106	Risk Management

Budget	Status
100000	Completed
50000	Ongoing
75000	Pending
90000	Onging
65000	Completed
80000	Pending

LEFT Join

2) Retrieve all employees and their assigned projects, including employees who have no projects using a LEFT JOIN.

```
SELECT A.'Employee ID', FirstName, LastName, Department,
        Salary, 'ProjectID', 'ProjectName', Budget, Status
FROM 'workspace'.'default'.'table_1_employee' AS A
LEFT JOIN 'workspace'.'default'.'table_2_project' AS B
ON A.Employee ID = B.Employee ID
```

EmployeeID	FirstName	LastName	Department	Salary	ProjectID	ProjectName
1	John	Doe	IT	70000	103	Cybersecurity Audit
2	Alice	Smith	HR	60000	102	Employee Training
3	Bob	Johnson	Finance	75000	104	Financial Analysis
4	David	Brown	IT	72000	null	null
5	Emma	Wilson	Sales	65000	105	Market Expansion
6	Michael	Clark	Finance	80000	106	Risk Management
1	John	Doe	IT	70000	101	AI Development

Budget	Status
75000	Pending
50000	Ongoing
90000	Ongoing
null	null
65000	Completed
100000	Completed

RIGHT Join

3 Retrieve all projects and their assigned employees, including projects that have no employees using a RIGHT JOIN.

```
SELECT B.`ProjectID`, ProjectName, Budget, Status,
        A.EmployeeID, `FirstName`, LastName, Department,
        Salary,
  FROM `workspace`, `default`, `table_1_employees` AS A
RIGHT JOIN `workspace`, `default`, `table_2_project AS B
ON A.EmployeeID = B.EmployeeID;
```

ProjectID	ProjectName	Budget	Status	EmployeeID	FirstName	LastName
101	AI Development	100000	Completed	1	John	Doe
102	Employee Training	50000	Ongoing	2	Alice	Smith
103	cybersecurity Audit	75000	Pending	1	John	Doe
104	Financial Analysis	90000	Ongoing	3	Bob	Johnson
105	Market Expansion	65000	Completed	5	Emma	Wilson
106	Risk Management	80000	Pending	6	Michael	Clark

Department	Salary
IT	70 000
HR	60000
IT	70000
Finance	75000
Sales	65000
Finance	80000

Full Outer Join

4) Retrieve all employees and projects, including those without a match in either table using a FULL JOIN.

```
SELECT A.`EmployeeID`, FirstName, LastName, Department, Salary,
        `ProjectID`, `ProjectName`, Budget, Status
FROM `workspace`, `default`, `table_1_employee AS A
FULL OUTER JOIN `workspace`, `default`, `table_2_project AS B
ON A.EmployeeID = B.EmployeeID;
```

Employee ID	First Name	Last Name	Department	Salary	Project ID
3	Bob	Johnson	Finance	75000	104
4	David	Brown	IT	72000	null
6	Michael	Clark	Finance	80000	106
2	Alice	Smith	HR	60000	102
5	Emma	Wilson	Sales	65000	105
1	John	Doe	IT	70000	103
1	John	Doe	IT	70000	101

Project Name	Budget	Status
Financial Analysis	90000	Ongoing
null	null	null
Risk Management	80000	Pending
Employee Training	50000	Ongoing
Market Expansion	65000	Completed
Cybersecurity Audit	75000	Pending
AI Development	100000	Completed

UNION

5) Retrieve a list of all unique cities where employees are located and project statuses.

→ Return: Location (Rename the column to Location using an alias).

```
SELECT Status AS Location
FROM 'workspace'.'default'.'table_2_projects
UNION
SELECT City AS Location
FROM 'workspace'.'default'.'table_1_employees';
```

Location
Completed
Ongoing
Pending
New York
Los Angeles
Toronto
London
Sydney

6) <u>UNION All</u>

Retrieve a list of all cities where employees are located and project statuses, allowing duplicates.

Return: Location (Rename the column to location using an alias)

```
SELECT Status AS Location
FROM 'workspace'.'default'.'table_2_projects
UNION All
SELECT City AS Location
FROM 'workspace'.'default'.'table_1_employees;
```

Location
Completed
Ongoing
Pending
Ongoing
Completed
Pending
New York
Los Angeles

Toronto
London
Sydney
New York

7) Retrieve employees who earn more then 70,000.
- Return : EmployeeID, FirstName, LastName, Department, Salary.

```
SELECT EmployeeID, FirstName, LastName, Department, Salary
FROM `workspace`.`default`.`table_1_employees
WHERE Salary > 70000;
```

EmployeeID	FirstName	LastName	Department	Salary
3	Bob	Johnson	Finance	75000
4	David	Brown	IT	72000
6	Michael	Clark	Finance	80000

8) Retrieve employees working in either IT or Finance departments.
Return : EmployeeID, FirstName, LastName, Department, Salary

```
SELECT EmployeeID, FirstName, LastName,
       Department, Salary
FROM `workspace`.`defauld`.`table_1_employees
WHERE Department IN('IT','Finance');
```

EmployeeID	FirstName	LastName	Department	Salary
1	John	Doe	IT	70000
3	Bob	Johnson	Finance	75000
4	David	Brown	IT	72000
6	Michael	Clark	Finance	80000

9) Retrieve projects that are not yet completed
- Return: ProjectID, ProjectName, Budget, Status

```
SELECT ProjectID,
       ProjectName,
       Budget,
       Status
FROM `workspace`.`default`.`table_2-projects`
WHERE Status != 'Completed';
```

ProjectID	ProjectName	Budget	Status
102	Employee Training	50000	Ongoing
103	Cybersecurity Audit	75000	Pending
104	Financial Analysis	90000	Ongoing
105	Risk Management	80000	Pending

10) Retrieve projects that have a budget greater than 70,000 and are not completed.
- Return: ProjectID, ProjectName, Budget, Status

```
SELECT ProjectID, ProjectName, Budget, Status
FROM `workspace`.`default`.`table_2-projects`
WHERE Budget > 70000 AND Status != 'Completed';
```

ProjectID	ProjectName	Budget	Status
103	Cybersecurity Audit	75000	Pending
104	Financial Analysis	90000	Ongoing
106	Risk Management	80000	Pending

11) Retrieve employees from New York or Toronto, ordered by salary in descending order.

- Return: EmployeeID, FirstName, LastName, Department, Salary, City

```
SELECT EmployeeID,
       FirstName,
       LastName,
       Department,
       Salary,
       City
FROM `workspace`.`default`.`table_1-employees`
WHERE City IN ('New York', 'Toronto')
ORDER BY Salary DESC;
```

EmployeeID	FirstName	LastName	Department	Salary	City
6	Michael	Clark	Finance	80000	New York
3	Bob	Johnson	Finance	75000	Toronto
1	John	Doe	IT	70000	New York

12) Retrieve the top 3 highest-paid employees.

- Return: EmployeeID, FirstName, LastName, Department, Salary

```
SELECT EmployeeID, FirstName, LastName, Department, Salary
FROM `workspace`.`default`.`table_1-employees`
ORDER BY Salary DESC
LIMIT 3;
```

EmployeeID	FirstName	LastName	Department	Salary
6	Michael	Clark	Finance	80000
3	Bob	Johnson	Finance	75 000
4	David	Brown	IT	72000

13) Find the total salary per department, sorted in descending order.
- Return: Department, TotalSalary (Rename SUM(Salary) as TotalSalary).

```
SELECT Department,
        SUM(Salary) AS TotalSalary
FROM `workspace`.`default`.`table_1_employees
GROUP BY Department
ORDER BY TotalSalary DESC;
```

Department	TotalSalary
Finance	155000
IT	142000
Sales	65000
HR	60000

14) Find the average salary per city, but only include cities where the average salary is greater than 65,000.
- Return: City, AverageSalary (Rename AVG(Salary) as AverageSalary).

```
SELECT City,
        AVG(Salary) AS AverageSalary
FROM `workspace`.`default`.`table_1_employees
GROUP BY City
HAVING AverageSalary > 65000;
```

City	AverageSalary
New York	75000
Toronto	75000
London	72000

15) Count the number of employees per department, including only departments with more than one employee.

- Return: Department, EmployeeCount (Rename COUNT(EmployeeID) as EmployeeCount).

```
SELECT Department,
        COUNT(EmployeeID) AS EmployeeCount
FROM `workspace`.`default`.`table_1_employees
GROUP BY Department
HAVING EmployeeCount > 1;
```

Department	EmployeeCount
IT	2
Finance	2

16) Retrieve the number of projects per status; but only include statuses with atleast 2 projects.

- Return: Status, ProjectCount (Rename COUNT(ProjectID) as ProjectCount).

```
SELECT Status,
        COUNT(ProjectID) AS ProjectCount
FROM `workspace`.`default`.`table_2_projects
GROUP BY Status
HAVING ProjectCount >= 2;
```

Status	ProjectCount
Completed	2
Ongoing	2
Pending	2

17) Retrieve the total project budget per employee, but only for employees who are managing projects worth more than 150,000.

- Return: EmployeeID, FirstName, LastName, TotalProjectBudget (Rename SUM(Budget) as TotalProjectBudget).

```
SELECT A.EmployeeID, A.FirstName, A.LastName,
        SUM(B.Budget) AS TotalProjectBudget
FROM `workspace`.`default`.`table_1_employees AS A
INNER JOIN `workspace`.`default`.`table_2_projects AS B
ON A.EmployeeID = B.EmployeeID
GROUP BY A.EmployeeID, A.FirstName, A.LastName
HAVING TotalProjectBudget > 150000;
```

EmployeeID	FirstName	LastName	TotalProjectBudget
1	John	Doe	175000